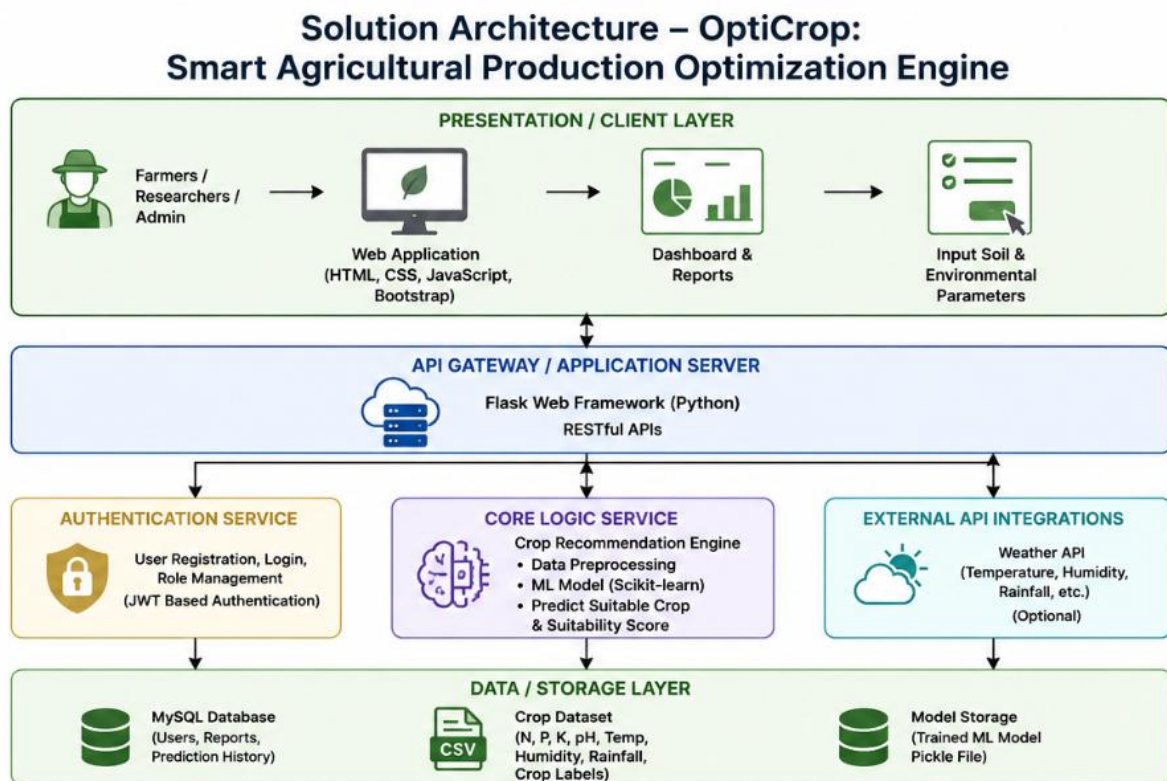


Solution Architecture

Date	29 June 2026
Team ID	xxxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table:

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides a user-friendly interface for farmers, researchers, and administrators to enter soil and environmental data and view crop recommendations.	HTML, CSS, JavaScript, Bootstrap
API Gateway / Application Server	Handles user requests, routes data between the frontend and backend, and manages application logic.	Flask (Python)
Core Logic Service	Processes soil parameters (N, P, K, pH, temperature, humidity, rainfall), runs the machine learning model, and generates crop recommendations.	Python, Scikit-learn, Pandas, NumPy
External API Integrations (Optional)	Retrieves weather information to improve crop recommendation accuracy.	Weather API (Optional)
Data / Storage Layer	Stores crop datasets, user details, prediction history, and system data.	MySQL, CSV Dataset